Event-Gated Ensemble Enhancement with ASR-Consistent Training for Indic Noise Event Removal

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Abstract—We present our Track 2 system for the Datathon@IndoML 2026 noise event removal challenge. The task is to suppress non-speech events (horns, animals, breathing, children, alarms, machinery, music) in real-world Indic speech from the Vaani corpus. The score is the SI-SDR on synthetic clips plus 100× the WER reduction of a frozen mandated ASR (SraVaani-1.0) over all clips. A single learned enhancer is applied identically to every clip. It averages a waveform-domain DNS64 (Demucs) model and a band-split RNN with a learned frame-level event gate, so the network detects where events are and leaves noise-free speech untouched. Both models are trained on on-the-fly mixtures whose event statistics follow the validation set, with an ASR-consistency loss on the frozen SraVaani encoder. A second stage fine-tunes them on longer, denser and louder mixtures, selecting checkpoints on a pseudo-test assembled from held-out validation clips. That pseudo-test predicted our leaderboard score within 0.13. Our final submission (Codabench 995085) reaches SI-SDR 12.78 dB and ΔWER +1.87 (synthetic +5.41), for a combined score of 14.65 (Phase 2 rank 13), with the hidden ASR audit passed.

I. Training Data and Preprocessing

Vaani Noise Event Dataset [1] (90,637 segments, 154.6 h, Gold/Silver/Bronze annotation tiers). We use it as a clean-speech pool. From Gold and Silver clips we keep non-event regions of at least 1.5 s that lie at least 0.25 s from any labelled event, and only from clips whose 10th-percentile frame energy is more than 28 dB below the clip RMS. This yields 8 h of low-background speech.

IndoML validation set (2,501 clips; 1,806 natural, 695 synthetic with clean references). The 695 synthetic pairs are split 560 for training and 135 for a holdout. From the 560 we take the clean references (0.81 h) and 1,783 noise events cut from noisy − clean at the labelled spans (±50 ms). In these pairs the added noise lies entirely inside the spans; outside them the noisy signal equals the clean one.

External data. We use ESC-50 [6] (2,000 environmental clips, resampled to 16 kHz and trimmed to the region within 30 dB of the peak) as a second noise source, so the model generalises beyond the validation noise. We also use Google FLEURS [7] Indic train shards (hi, bn, te, ta, kn, ml, mr, or, as, gu, pa, ne; at most 600 utterances per language, 12 s or shorter) as additional clean read speech. In total the banks hold 3,783 noise events (2.72 h) and up to 17.6 h of clean speech, stored as memory-mapped float16 arrays.

Augmentation pipeline (on the fly; every batch is new).

1) Clean speech is a validation reference with probability 0.5, otherwise a Vaani or FLEURS segment. Utterances, each rescaled within ±3 dB and separated by 50–600 ms gaps, are joined until the 6–10 s crop is full. The crop is scaled to $-30 \ldots -15$ dBFS.
2) The number of events is $k = n \cdot U(1, L/4.5\,\text{s}) \cdot U(\text{ev_mult})$, where n is drawn from the per-clip event counts of the validation synthetic set.
3) Each event is a validation noise event (p=0.7) or an ESC-50 crop of empirical event duration (p=0.3) with 5 ms fades and random polarity, placed at a uniform offset. Its level relative to the clean RMS is drawn from the empirical validation distribution (median +4.3 dB), plus $\mathcal{N}(0, 2)$ dB, plus $U(\text{db_shift})$.
4) The frame event mask (hop 128) is the target of the activity head. Mixture and target are peak-normalised jointly.

The simulated mixtures match validation (median input SI-SDR −2.5 vs. −2.6 dB; event coverage 43% vs. 45%). Stage 2 uses ev_mult = (1, 2), db_shift = (0, 3) dB and 10 s crops.

II. Proposed Method

A. Models

DNS64. The Demucs waveform U-Net with an LSTM bottleneck [2] (hidden width 64, 33.5 M parameters), initialised from the public checkpoint dns64-a7761ff99a7d5bb6.th (facebookresearch/denoiser).

Event-gated BSRNN (ours, 11.3 M parameters), after band-split RNN [3].

- A 512-point STFT (hop 128) is split into 37 bands of 16×4, 16×8, 4×16 and 1 bins. Each band is normalised and projected to 112 features.
- Six blocks alternate residual BLSTMs over time and over bands. Per-band GLU MLPs output a complex mask $\mathbf{M}$, so $\hat{s} = \text{iSTFT}(\mathbf{M} \odot \mathbf{X})$.

- A gate head maps the concatenated band features to an event probability $p(t)$. The output is

$$y = x + g(t)\,(\hat{s} - x), \qquad g(t) = \gamma + (1-\gamma)\,p(t), \quad (1)$$

with floor $\gamma = 0.2$, so every frame is still processed while event-free speech stays close to the input.

B. Losses

For mixture x, clean target s and event mask m,

$$\mathcal{L}_{\text{BSRNN}} = \tfrac{1}{2}\mathcal{L}_{\text{rec}}(\hat{s}, s) + \tfrac{1}{2}\mathcal{L}_{\text{rec}}(y, s) + \text{BCE}(p, m) + \lambda\,\mathcal{L}_{\text{ASR}}, \quad (2)$$

$$\mathcal{L}_{\text{rec}}(a, s) = -\,\text{SI-SDR}(a, s) + 0.5\,\text{MRSTFT}(a, s), \quad (3)$$

and $\mathcal{L}_{\text{DNS64}} = \mathcal{L}_{\text{rec}}(\hat{s}, s) + \lambda\mathcal{L}_{\text{ASR}}$.

- MRSTFT is the L1 distance between log-magnitudes at FFT sizes 256, 512 and 1024 [5].
- SI-SDR is the metric itself [4].
- The ASR-consistency loss $\mathcal{L}_{\text{ASR}}$ is the L1 distance between the frozen SraVaani-1.0 encoder [8] outputs on the enhanced and clean signals, normalised by the target's mean magnitude. It is applied every second step, with $\lambda = 0.1$ (0.2 for one DNS64 run). Its front end replicates the ASR's pre-emphasis, 128-bin log-mel and per-feature normalisation. Gradients flow through the frozen encoder into the enhancer, so the enhancer learns outputs the mandated ASR transcribes like clean speech.

C. Training strategy

Optimiser. AdamW [10] (weight decay 10^{-2}) with a 500-step warm-up and a cosine schedule driven by wall-clock time, which guarantees completion inside a fixed GPU allocation. We use gradient clipping at 5 and bf16 autocast for BSRNN, and keep the best checkpoint by evaluation SI-SDR.

Stage 1 (holdout selection):

- DNS64: 6 s crops, learning rate 10^{-4}, 210 min on an NVIDIA B200.
- BSRNN: trained from scratch, 6 s crops, learning rate 10^{-3}, 180 min on an A100 80 GB.
- A second DNS64: 8 s crops, learning rate $2{\cdot}10^{-4}$, $\lambda = 0.2$, 150 min on an A100.

Pseudo-test. After the first submission showed a large gap between validation and test, we built a pseudo-test only from the 135 holdout clips. It has 106 clips of 2–3 joined holdout references (median 13.5 s) carrying their own noise plus 0.5× extra holdout events at +1.5 dB. Its median input SI-SDR is −4.5 dB, which matches the noise level we estimated for the real synthetic test clips (−4.48 dB). It reproduces the real per-clip score distribution of our first submission (median 12.79 vs. 12.27 dB; 5th percentile 4.78 vs. 4.59 dB).

Stage 2 fine-tunes DNS64 and BSRNN on test-like mixtures for 50 min each on the B200. Checkpoints are selected on the pseudo-test, and BSRNN's clean pool excludes the Vaani segments.

TABLE I
SI-SDR (dB) on the validation holdout (135 clips) and the pseudo-test (106 clips), and the Phase 2 test score (SI-SDR / ΔWER / combined).

System	Holdout	Pseudo-test	Test (Codabench)
Noisy input	−0.76	−3.75	–
DNS64 (stage 1)	19.04	12.81	–
BSRNN, gated (stage 1)	19.04	12.67	–
DNS64 + BSRNN	19.78	13.36	12.66 / 1.80 / 14.46
+ second DNS64	19.71	13.36	– / – / 14.41
Stage-2 DNS64 + BSRNN	–	13.52	12.78 / 1.87 / 14.65

Inference. The final system averages the two stage-2 models, with gate floor 0.2. Output RMS is matched to the input, clips are processed in 20 s chunks with a 2 s cross-fade, and audio is written as PCM-16. Transcripts come from the unmodified SraVaani-1.0 (fp32, greedy TDT [9]) on the written WAVs.

Candidate selection. Candidates are ranked by an estimated leaderboard score:

- pseudo-test SI-SDR minus a 0.70 dB calibration offset;
- plus 0.32× the synthetic ΔWER, measured on holdout clips against ASR-on-clean transcripts and calibrated to our first submission;
- plus 0.68× the natural-validation ΔWER.

The weights 0.32 and 0.68 are the synthetic and natural shares of words in the test set.

III. Results and Conclusion

Table I summarises the results.

- Final submission (995085): SI-SDR 12.78 dB and ΔWER +1.87, of which +5.41 on synthetic and +0.21 on natural clips. Combined 14.65; the hidden ASR audit passed with 0% mismatch.
- Architectural diversity helps. DNS64 + BSRNN gains 0.75 dB over either model alone and turns the holdout ΔWER positive, whereas averaging two DNS64 models adds nothing.
- The learned gate is only useful with good data. BSRNN gated at $\gamma = 0.2$ reaches 19.04 dB, against 17.82 dB ungated. In a run whose clean pool also contained noisier Vaani speech, the gate instead lowered SI-SDR (10.5 vs. 16.5 dB ungated), consistent with activity targets that mislabel real background noise as speech.
- Noise level, not duration, limits performance. On the scored test clips, per-clip SI-SDR correlates 0.82 with the estimated input SNR and 0.09 with clip length. Clips below −6 dB input SNR (39%) average 8.5 dB, and clips above 0 dB average 20.7 dB. On the pseudo-test our ensemble (13.36 dB) already exceeds the ideal ratio mask (12.99 dB) and approaches the ideal phase-sensitive mask (15.15 dB): mask-style enhancement has little headroom left at such SNRs.

- Pseudo-test fidelity. The estimator predicted 14.52 for our final entry, which scored 14.65. Validation-holdout estimates were optimistic by about 5 dB, because test clips are 2.7× longer and much noisier.

Novelty. Our contributions are (i) an event-gated BSRNN that learns event activity jointly with separation; (ii) ASR-consistency training through the frozen mandated ASR; (iii) wall-clock-scheduled training with validation-matched on-the-fly mixing; and (iv) a holdout-derived pseudo-test together with a calibrated estimate of the leaderboard score, used to choose submissions.

Conclusion. A single learned, event-aware enhancer trained on statistically matched mixtures delivers strong SI-SDR, with a positive ASR gain on both natural and synthetic speech. The remaining gap lies in very low-SNR clips, where stronger separation models and longer test-condition training are the clearest next steps.

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